

Functional Features Audit & Evaluation Matrix

Ecosystem Product Scope Checklist • Developer: **Bhargavasai**

Evaluation Date	15 March 2024
Team Identification ID	XXXXXX
Project Assignment Name	Credit Card Fraud Detection System using Machine Learning
Maximum Evaluation Marks	5 Marks

1. SOLUTION FUNCTIONAL FEATURES MAPPING

Comprehensive catalog listing operational project capabilities, software module components, and targeted rubric weight evaluations.

S.NO	FEATURE NAME	FEATURE DESCRIPTION	MODULE / COMPONENT	STATUS	MARKS WEIGHT
1	User Input Interface	Responsive text form elements capturing structural transaction dimensions (Time, Amount, V1-V28 vectors) cleanly from the client web console.	predict.html (UI)	Done	Core / High
2	Machine Learning Pipeline	Optimized classification weights saved and serialized via single-cycle Joblib configurations for precise, rapid inference calls.	fraud_detection_model.pkl	Done	Core / High
3	Real-Time Prediction Engine	Backend operational routing endpoints intercepted by the Flask application framework to manage vector formatting and decision thresholds.	app.py (Backend)	Done	Core / High
4	Dynamic Results Visualizer	Instantly maps runtime inference values to display responsive alert banners stating explicitly "Normal Transaction" or "Fraud Transaction Detected."	result.html (UI)	Done	Core / Medium
5	Automated EDA Visualizations	Integrated plots mapped to the static image asset directories showcasing value distributions, anomalies, and multi-variable correlation heatmaps.	static/img/ (Visuals)	Done	Core / Medium
6	Dynamic Port Management	Runtime configuration script automatically catching socket address assignments and redirecting fallback routing if standard ports are occupied.	app.py (Core Logic)	Done	Nice-to-Have / Low
7	Project Home Page	Introductory interface detailing automation benefits, underlying dataset scopes, engineering milestones, and live workspace pipelines.	index.html (UI)	Done	Core / Medium

2. OPERATIONAL METRICS & STRUCTURAL BREAKDOWN

Feature Lifecycle Summary	
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	7 (Input UI, ML Engine, Preprocessing, Results Banner, EDA Visuals, Home UI, Navigation Grid)
Nice-to-Have / Fallbacks	1 (Dynamic Port Architecture Fallback)
Verified & Tested Blocks	8

S.No	Category	Features in Category / Examples
1	User Interface (UI)	Landing Page (`index.html`), Form Data Control (`predict.html`), Results Display (`result.html`), Master Boilerplate (`base.html`).
2	Backend / Logic	Real-time prediction pathways (`app.py`), dynamic runtime network port validation routines.
3	Storage / Models	Serialized machine learning weight pipeline binaries (`fraud_detection_model.pkl`).
4	API / Integration	Secure data passage layers linking HTML presentation models to Python array processing structures.
5	Security / Validation	Payload boundary sanitization and rigorous constraint checks on numerical form inputs.